

Product	Name	ND-no.	Category
	COSMOS Ethernet Option	210580B	STPR
Reason	. New product . Error Correction X Change/Addition . Different Environment		
Documentation	Title	ND-no.	
	COSMOS Operator Guide	30.025.2 EN	
Purpose	To allow use of Ethernet (ISO DIS 8802/3 CSMA/CD) as transmission medium under COSMOS as the software part of the following products: - ND-no. 110063 Ethernet Controller (new, 1-board) - ND-no. 108630 Ethernet Controller (old, 2-board)		
Prerequisites	Computer Type Floating format Op. system Version		
	ND-100 All SIN III >= J6100		
	ND-500 All SIN III >= J6100		
	NOT ND-Satellite		
	ND-no. Product name		
	210373J X-MESSAGE		
	210374D COSMOS Basic Module		
	210721A BRF-LINKER for ND-100		
	SINTRAN III J must be generated with PIOC datafield.		
	SINTRAN III K must be generated with Ethernet Interface datafield.		
	Minimum mass storage resources for installation		
	User	Userspace	Number of files
	SYSTEM	120 pages on	2 files
	Any user (UTILITY)	340 pages on	8 file
	Minimum permanent mass storage resources		
	User	Userspace	Number of files
	Any user (UTILITY)	340 pages on	8 files
	Max/Min number of RT-descriptions: Max 2 Min 2		
	Max/Min number of segments: Max 6 Min 3		
	Space requirements on segment file(s) is 303/87 pages		
		ND-no. for Source	
		210581B	

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File Name	Type	Containing
ENCOS-SER-I-B<rev>	DSEG	Server for 108630
ENCOS-ERR-I-B<rev>	BRF	Supervisor for 108630
ENCOS-MON-I-B<rev>	PROG	Monitor for 108630
ENCOS-SER-B0-B<rev>	BPUN	Server for 110063, bank 0
ENCOS-SER-B1-B<rev>	BPUN	Server for 110063, bank 1
ENCOS-SER-B2-B<rev>	BPUN	Server for 110063, bank 2
ENCOS-SER-B3-B<rev>	BPUN	Server for 110063, bank 3
ENCOS-ERR-II-B<rev>	BRF	Supervisor for 110063
ENCOS-MON-II-B<rev>	PROG	Monitor for 110063
ENCOS-IN-B<rev>	PROG	Installation program
PO-PWRFAIL-A<rev>	PROG	Powerfail handler
UE-ERMSG-EN-B03	ERR	Error message database

NOTE: <rev> is to be replaced by the current version of the DIRECTORY or FILE. The revision level is found on the preceding "ND SOFTWARE LIBRARY DISKETTE" pages.

1 NEW FEATURES

- New hardware supported.
- Faster asynchronous communication between ND-100 and Ethernet Controller for XMSG version L on SINTRAN VSX.
- Trace and statistics dump after fatal errors.
- Powerfail handling.
- General performance and resilience improvements.

2 INSTALLATION

Log in as user SYSTEM. If one version of COSMOS Ethernet Option is currently running on your system, then it must be stopped prior to running this installation job.

Initial installation

Insert the (first) installation floppy and enter the directory. Start the installation program by typing:

```
@(210580:FLOPPY-USER)ENCOS-INJ
```

This program will copy the files from floppy and generate a COSMOS Ethernet server. For operator interaction with the installation program please refer to the example on the next page.

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```

=====
*      COSMOS Ethernet Option version B01 - 21.01.87      *
*      INSTALLATION PROGRAM                                *
*-----*
*      This program loads software for :                  *
*-----*
*      - ND 108630 Ethernet Controller (old,2-board version) *
*      - ND 110063 Ethernet Controller (new,1-board version) *
*-----*
*      Space needed for installation :                     *
*-----*
*      - On any user selected below 310 available pages.   *
*      - On user SYSTEM while installing 110 available pages.*
*-----*
*      General information :                               *
*-----*
*      - The software will give best performance with XMSG VSX L. *
*      - Reinstallation is necessary, when changing from      *
*      SINTRAN Version J to K (or later).                     *
*      - If your system is ordered with RTCOMMON, please remember *
*      to remove it from the Ethernet Controller memory.      *
*      - You may stop the installation at any time by typing ESC. *
*-----*
>> User for COSMOS Ethernet files (Default is UTILITY):J
>> Specify Ethernet Controller type(old/new)(Def. new):J (ONLY SIN J)
>> Specify Ethernet Interface number (Default is 0):J
--> Double density floppy.
--> Copying the installation program to user SYSTEM.
--> Copying the error message database to user SYSTEM.
--> File was not copied as newer version exists.
--> Searching for the subsystem BRF-LINKER
--> Checking Ethernet Controller in SINTRAN.
--> ND 110063 Ethernet Controller (new,1-board) found. (NOT SIN J)
--> Copying ENCOS-ERR-II-B01:BRF
--> .....
--> .....
--> Copying ENCOS-SER-B3-B01:BPUN
--> Copying the powerfail handler to user UTILITY.
--> Configuring the supervisor program. Wait...
--> Creating load file. Wait...
--> Executing load file. Wait...
=====
*      COSMOS Ethernet Option B01 is now installed.      *
*-----*
*      - For starting and configuring the software, please refer *
*      to the PD-sheets and the COSMOS Operator Guide.      *
*      - If any trouble arises starting COSMOS Ethernet Option *
*      please inspect the ENCOS-LOAD-0-B01:LIST-file on SYSTEM.*
*      - Please include the (UTILITY)ENCOS-LOAD-0-B01:MODE-file *
*      in the COLD-START procedure.                        *
*-----*
=====

```

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The Ethernet Interface number refers to the setting of the device number thumbwheel on the Ethernet Interface board. (The Ethernet Master board if old Ethernet Interface.)

It is important to keep the the list-file from the installation mode-job in case any abnormal behavior should occur later. For example, by copying the list file to the user where the reset of the COSMOS Ethernet Option files are stored.

Later installations

When upgrading from SINTRAN J to SINTRAN K on your system you should do a new installation as the installation program does some checking that is not performed by the mode file mentioned below. The procedure is the same as for the initial installation.

When upgrading from the old Ethernet Interface (ND-108630) to the new Ethernet Interface (ND-110063) a new installation of the software must be performed.

Cold start

The installation program generates a mode file that should be included in the HENT mode file to be run after a cold start.

3 OPERATOR COMMANDS

The COSMOS Ethernet Option uses only a subset of those XMSG-COMMAND directives used by the X21 and X25 options. The specific use of the commands for COSMOS Ethernet Option will be included in the COSMOS network operator documentation, but is also explained below.

The COSMOS Ethernet servers are named by the following convention :

```
ENNSi ==> EtherNet Network Server i.
           Where i is the same number as the thumbwheel setting
           of the Ethernet Interface.
```

The "Server system name" parameter accepts carriage return as local system.

EXAMPLE

COMMENTS

```
X-C:START-NETWORK-SERVER↓
Server name?ENNS0↓
Server system name?↓      => Default is local system
Window size?↓             => Number of receive buffers (Def.5)
Wide Area Network(Y/N)?N↓ => Local Area Network
```

NOTE: Maximum number of receive buffers for 108630-version is 6.

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X-C:STOP-NETWORK-SERVER↓

Server name?ENNSO↓

Server system name?↓

=> Default is local system.

X-C:DEFINE-NETWORK-CONNECTION↓

Remote system?ND-548↓

=> Earlier defined system name.

Server name?ENNSO↓

=> Last used server is default.

Server system name?↓

=> Default is local system.

Remote DTE address?0↓

Dial-up cost?0↓

=> [Meaningless
on LAN,
but default is
not accepted.

Connect cost/minute?0↓

Data cost/ksegment?0↓

Other useful general commands are:

X-C:LIST-ROUTES,,ND-548↓X-C:LIST-NETWORK-SERVERS↓

Name	Sysid	Link	Network	Xnser-port-Xgate	Rcv-buff-Xmit
ENNSO	9800	2	LAN	3 4	5 2

For XMSG version K or later only:

X-C:LIST-VERSION,,↓

System	Version	Patch	Product number/name
XMSG	K01	3	210373 / X-message for SINTRAN III/VSX
ENNSO	B01		210580 / COSMOS Ethernet I - IOC Server

COSMOS Ethernet Option may identify itself with four different texts depending on Controller, XMSG and SINTRAN versions.

Controller	SINTRAN	XMSG	Text
ND-108630	J	J	Ethernet I - XMSG Server
ND-108630	K	K	Ethernet I - XMSG Server
ND-108630	K VSE	L	Ethernet I - XMSG Server
ND-108630	K VSX	L	Ethernet I - IOC Server
ND-110063	J	J	Ethernet II - XMSG Server
ND-110063	K	K	Ethernet II - XMSG Server
ND-110063	K VSE	L	Ethernet II - XMSG Server
ND-110063	K VSX	L	Ethernet II - IOC Server

For further information please refer to the COSMOS Operator Guide.

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The Ethernet server startup sequence should be included in the WARM-START-file (LOAD-MODE) after the XMSG-STARTUP mode file. Please refer to the example below:

```
@ABORT ENNSO
@RT ENNSO
@HOLD 0,0
@HOLD 5,2
@(UTILITY)XMSG-COMMAND
SET-PRIVILEGED
'
' All mapping between COSMOS names and System numbers defined in
' the XMSG-STARTUP file.(DEFINE-REMOTE-NAME COMMANDS)
'
' Start the server and declare the
' systems that are connected to the Ethernet.
'
START-NETWORK-SERVER ENNSO,,,N
DEF-NETWORK-CONN OLE ENNSO,,,0,0,0,0
DEF-NETWORK-CONN DOLE ENNSO,,,0,0,0,0
DEF-NETWORK-CONN DOFFEN ENNSO,,,0,0,0,0
DEF-NETWORK-CONN SKRUE ENNSO,,,0,0,0,0
DEF-NETWORK-CONN KLARA ENNSO,,,0,0,0,0
DEF-NETWORK-CONN ANTON ENNSO,,,0,0,0,0
DEF-NETWORK-CONN MIKKE ENNSO,,,0,0,0,0
'
'Then have a look at the status.
'
LIST-NETWORK-SERVERS,,
LIST-VERSIONS
EXIT
@CC ***** END *****
```

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4 SPECIAL CONSIDERATIONS FOR OLDER XMSG VERSIONS

If installing the ND-110063 Ethernet Interface on a system with XMSG K (or older) or on any SINTRAN VSE system, the following change must be done to the default XMSG generation variables:

Call the XMSG-installation program:

```
@(UTILITY)XCOM (UTILITY)XMSG-INJ (XMSG J)
```

or

```
@(UTILITY)XMSG-INJ (XMSG > J)
```

```
> Which user has the X-message files (default:UTILITY):J
```

```
..... MENU ON SCREEN .....
```

```
> Please enter ==>3J (Redefine XMSG generation variables.)
```

```
*** PLEASE WAIT ***
```

```
.. An information header and the first parameter is displayed ..
```

```
.. Type <CR> until .....
```

```
> X4MCB, Number of calls in a multical function(oid:5):8J
```

```
.. Some other parameter ...J
```

```
.. The new parameter set is displayed..
```

```
> Type "CR" to get back to main menu. ==>J
```

```
..... MENU ON SCREEN .....
```

```
> Please enter ==>8J (Exit)
```

```
The reload xmsg: @(utility)xmsg-load:modeJ
```

In addition to the change above, one change should be done to X-MESSAGE version J (ND-210373J), prior to COSMOS Ethernet Option installation. Same procedure as above, but change the below variable:

```
...
```

```
.. Type <CR> until .....
```

```
> X4ACK, Number of network ACK frames (oid:5):20J
```

```
...
```

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5 TRACE AND STATISTICS FACILITIES

When the network server stops, either due to a normal command (SINTRAN-SERVICE; stop-xmsg or XMSG-COMMAND; stop-network-server) or due to a fault, the system statistics for the Controller will be dumped to a file (RT)ENCOS-DUMP:DATA. If trace is enabled, the contents of the trace buffer will also be dumped to this file.

NOTE : The next time the network server is stopped, the dump-file will be overwritten.

A monitor program is included in order to enable the experienced user, ND-service and ND-support to locate errors, get indications of network overload etc. The program is called by:

@(utility)encos-mon↓

...

ENTER COMMAND:

Enable/Disable trace

ENTER COMMAND:trac↓

manipulate trace

server number (0/1/2/3):↓ => 0 is default

function enable/disable/read (E/D/R)?E or D↓

<enable for specific remote system only(Y/N)?↓>

Dump trace

ENTER COMMAND:trac↓

manipulate trace

server number (0/1/2/3):↓ => 0 is default

function enable/disable/read (E/D/R)?R↓

filter display by system number (Y/N)?N↓

via physical copy (Y/N)?

output to file (ENCOS-TRACE:DATA)(Y/N)?

Copying.....

Formatting to file...

The trace can now be inspected by reading the file into any editor.

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Get media-access or network level statistics

```

ENTER COMMAND:stat↓
get statistics
server number (0/1/2/3):↓    => 0 is default
server system name:↓        => local system default
source of statistics is specified by integer code as follows:
0  => Media Access statistics
N  => Network level statistics for connection to system N
system statistics source (0/N):0 or N↓
...

```

Note that even though the monitor program comes in two versions for new and old Ethernet Interfaces, remote statistics may be obtained for both types of Ethernet Interfaces from both versions of the monitor.

6 POWERFAIL HANDLING

The powerfail handler, POPWR, is an RT program that logs in on a Terminal Access Device on the local computer. POPWR will execute any mode file that does not stop TADs or XMSG. The hardcoded input- and output-file names are:

(SYSTEM)PO-PWRFAIL:MODE and (SYSTEM)PO-PWRFAIL:OUTP

The last command executed by the TAD is 'LOGOUT'. When this command is performed, POPWR will go into passive mode. To enable the program to be started at powerfail one must enter SINTRAN-SERVICE-PROGRAM and type:

```

DEFINE-USER-RESTART-PROGRAM POPWR
or (SINTRAN Version K only)
NEXT-USER-RESTART-PROGRAM POPWR if the command
LIST-USER-RESTART-PROGRAM gives any RT program name as output.

```

Execution of the Ethernet startup mode file, (Please refer to earlier given example), and any other mode files for devices that do not handle powerfail are then included in the PO-PWRFAIL:MODE file.

NOTE: SINTRAN versions older than K may only have ONE user restart program.